

Data Visualization For Business Decisions

**A Framework For Analysis and Improvement
of Data Visuals**

Dr. Andres Fortino

Table of Contents

INTRODUCTION3

THE ANALYSIS FRAMEWORK.....4

CHAPTER 1.....11

CHAPTER 2.....15

CHAPTER 3.....18

CHAPTER 4.....21

EXERCISE 5.....24

EXERCISE 6.....27

REFERENCES.....30

Introduction

This workbook was written for business analysts who wish to increase their skills in improving data visuals and creating compelling presentations used to support business decisions. It is a qualitative lab to develop the power of visualization and discrimination. It does not require the reader to modify charts but to analyze and describe what would improve charts. You will to use Excel to run the scoresheet.

In a set of controlled exercises, the reader is taken through the eighteen elements of six dimensions of analyzing and improving a chart or visual used to communicate a business point. An analyst, or anyone analyzing data, would typically create visuals of the results as the analysis goes along. These are graphs of data for analysis; they are rough and often no thought is given to making them compelling at that point. Probably no one other than the analyst will ever see them. These graphs may even accumulate in an electroclinic research notebook (typically a PowerPoint document) with slides. At the end of the analysis, these graphs, together with numerical summaries of results, are used by the analyst to come to certain conclusions.

Then comes the time to communicate. Often, they are not given a lot of time to present their findings. This is where the work of neurobiologist John Medina comes in to play. He exhorts us to use no more than ten minutes to make our case, lest we bore our audience to inattention. In any event, we must present our findings with as few slides as possible. The analysis looks over the rough graphs produced in analysis, look at the conclusions as then asks: "which of these are the most powerful visuals to make the point and underscore conclusions in the most compelling way?" There are probably no more than three or four such visuals that have to be created.

The next step is to create those compelling visuals that tell the story. That's what these exercises help you do: refine your skills in turning a rough graph into a compelling visual. One way to do that is to refine your "seeing eye." Can you see what is wrong? What must be done better? How do you improve a chart to make it more compelling? Many issues are very subtle so it takes work to develop a reined sense of sight. Do these exercises often and the process will become second nature. As you work through these exercises you will internalize the observe-analyze-refine process to improve your visuals.

Dr. Andres Fortino
December 2024

The Analysis Framework



Data visualization is a critical skill in today's data-driven world. The ability to create clear, compelling, and informative visual representations of data can make the difference between insights being understood and acted upon, or lost in a sea of numbers. This chapter introduces a comprehensive framework for analyzing and improving data visualizations, built around an 18-point checklist organized along six key dimensions.

The six dimensions - Story, Signs, Purpose, Perception, Method, and Charts - provide a holistic approach to evaluating and enhancing data visuals. Each dimension focuses on specific aspects of effective visualization:

1. Story: Ensuring the visual tells a clear narrative
2. Signs: Using appropriate symbols and maintaining a high signal-to-noise ratio
3. Purpose: Aligning with organizational needs and audience expectations
4. Perception: Leveraging principles of visual perception for clarity
5. Method: Employing best practices in color use, decluttering, and titling
6. Charts: Selecting and implementing the most appropriate chart types

Within these dimensions, the 18-point checklist serves as a practical tool for assessing and refining data visualizations. It prompts creators to consider crucial elements such as the clarity of the visual story, the effectiveness of signs and symbols, the match between the visualization and its intended purpose, the application of perceptual principles, the judicious use of visual methods, and the selection of appropriate chart types.

The chapter demonstrates how to apply this checklist through a series of case analyses. Each case focuses on one dimension, guiding readers through the process of critically examining a visualization and identifying areas for improvement. This step-by-step approach allows readers to internalize the principles and develop their analytical skills progressively.

Traditionally, applying such a checklist would require significant time and expertise from a human analyst. However, the chapter also explores an exciting new frontier: the use of large language models (LLMs) in data visualization analysis. By fine-tuning an LLM with the principles embodied in the 18-point checklist, we create a powerful AI assistant capable of rapidly analyzing and suggesting improvements for data visualizations.

The chapter concludes by comparing the manual analysis process with the AI-assisted approach. This comparison highlights both the progress made in automating complex analytical tasks and the continued importance of human judgment and creativity in data visualization. While the AI can quickly identify potential issues and suggest improvements based on established principles, the human analyst brings contextual understanding, creative

problem-solving, and the ability to make nuanced decisions that go beyond rule-based assessments.

By presenting both traditional and AI-assisted methods, this chapter equips readers with a versatile toolkit for creating and improving data visualizations. Whether working through the checklist manually to develop a deep understanding of visualization principles, or leveraging AI to streamline the analysis process, readers will gain valuable skills for communicating data effectively in various professional contexts.

As we navigate through the chapter, readers will not only learn how to critique and enhance existing visualizations but also how to apply these principles in creating their own compelling data stories. This comprehensive approach to data visualization will empower professionals to transform raw data into impactful visual narratives that drive understanding and decision-making.

An Analysis Tool for Visuals

We will use an analysis template provided with this manual to analyze and improve a chart, or data visual, along the six dimensions outlined above and discussed in detail at the start of each lab. We are not trying to get the analysis perfect or use the template with a great degree of accuracy. It is a tool to guide the eye and your discrimination. Using the analysis template and the associated questions produces an imperfect indicator and it should be sufficient to guide you in making improvements to any chart. We hope that by using the tool multiple times you will then begin to internalize the questions and eventually you will not need the tool.

To use the tool, consider answering the question associated with each principle. Consider each question and then use the following rough criteria: answer YES if the chart fulfills the question for the most part (>65%), and NO if the chart is deficient in that question (<65%). This process requires a rough pass-fail judgment on the issue. It's not perfect and it is not meant to create an exact measurement of the visual's perfection. It is meant to develop and refine your power of discrimination when analyzing and improving charts.

The next few pages provide a copy of the analysis template to print as reference. You can copy them and use paper and pencil if you wish. Or do it electronically using your favorite editor. Use a different template for each chart being analyzed. You can use the template as provided right in this book.

Data Visualization for Business Decisions

Analyzing A Visual Analysis Template

Story

Visual Story - A picture is worth a thousand words

- ☐ *Is the point of the visual very clear?*

Visual Props - Visuals support the story, they are not the story

- ☐ *Has the visual has been simplified and focused?*

Storytellers - Standing on the shoulder of giants

- ☐ *Are past masters and the basic charts that they pioneered emulated?*

Analysis of Story

- ☐ *What is wrong with the visual being analyzed along this dimension?*
- ☐ *What can be done to improve the visual along this dimension?*

Signs

Signs - Bertain: Semiotics, sign, and signifiers

- ☐ *Is the use of signs and symbols appropriate?*

Communication - Visualization as a communication system

- ☐ *Is the Signal to Noise ratio high?*

Function - Cairo: The Functional Art - Art vs. Clarity

- ☐ *Is the chart functionally informational rather than beautiful art?*

Analysis of Signs

- ☐ *What is wrong with the visual being analyzed along this dimension?*
- ☐ *What can be done to improve the visual along this dimension?*

Purpose

Need- Needs of the requester and the organization

- ☐ *Does the chart fulfill organizational information needs?*

Audience - Cairo: Audience needs, biases, and journey

- ☐ *Does the chart allow for audience biases, needs, and journey?*

Frame - Shron: Thinking with Data CoNVO model

- ☐ *Does the visual answers a well-framed analytical question?*

Analysis of Purpose

- ☐ *What is wrong with the visual being analyzed along this dimension?*
- ☐ *What can be done to improve the visual along this dimension?*

Perception

Seeing – Anatomy: the human eye-brain system

- ☐ *Does the eye of the viewer focus on the most important point being made?*

Mind - Gestalt: psychology of perception

- ☐ *Have the principles of the Gestalt psychology of perception been thoughtfully employed in the visual?*

Quality - Alexander: Quality Without a Name

- ☐ *Does the visual inform the viewer and dispel his ignorance?*

Analysis of Perception

- ☐ *What is wrong with the visual being analyzed along this dimension?*
- ☐ *What can be done to improve the visual along this dimension?*

Method

Color - Use of color, dysfunctions

- ☐ *Is color used judiciously and sparsely?*

Chart Junk - Tufte: Chartjunk

- ☐ *Is the visual clear of unnecessary visual elements not leading to a clear point being made?*

Title – McKinsey: A Better Way to Title Charts

- ☐ *Does the title of the chart convey the point being made with the chart?*

Analysis of Method

- ☐ *What is wrong with the visual being analyzed along this dimension?*
- ☐ *What can be done to improve the visual along this dimension?*

Charts

Right Chart - Cleveland and McGill: Functional Scale

- ☐ Does the type of chart being used match the level of judgment required?

Selection – The Basic Charts

- ☐ Does the chart type used match the business question being answered

Tables - Analyze or Tell?

- ☐ Are referenceable visuals (tables) readable with appropriate conditional formatting and thumbnail graphs used for emphasis?

Analysis of Charts

- ☐ What is wrong with the visual being analyzed along this dimension?
- ☐ What can be done to improve *the visual* along this dimension?

Chapter 1

Story



Visual Story Telling

Al Shallowly, an expert visual storyteller, tells us that: *“Visualizations act as a campfire around which we gather to tell stories.”* Are we telling compelling business stories with our visuals?

Create a Visual Story

As the saying goes, a picture is worth a thousand words. Does your visual make a clear point that would take a lot of words to convey? What is the point of the visual? Does your audience get the point? Is your story clear?

Rather than use 1,000 words to describe the point of your data – show it! We will assume that you are working with business issues. You have data that points to the direction managers should take on the issues and support certain actions to take. Rather than describe the facts in words, draw a picture– not just by presenting the data. Do not “Let the data speak for itself,” but rather “Tell your story”. Draw a picture. The point of the visual is to make a business point. The data-driven visual has to convey the point unambiguously

Usually use one point per visual, or several, if in a dashboard. The point we are trying to make is a business point. It must be tied to business metrics. It is most effective when tied to fundamental metrics like KPIs (Key Performance Indicators) and CSFs (Critical Success Factors) that have been adopted by the organization. Visuals are always most effective when dealing with fundamental financial implications.

The point of this principle is that for a data visual to be most successful your visual should make a very clear business point - that is the story you are telling.

Visual Story - A picture is worth a thousand words

- ☐ Does the visual make a clear point that would take a lot of words to convey?
- ☐ Is the point of the visual clear?
- ☐ Do I easily get the point?
- ☐ Is the story clear?
- ☐ Does the visual support a business story, not a story of your analysis journey?

Is the point of the visual very clear?

Make It a Prop

Famous chart maker Stephen Few admonishes us that “Numbers have an important story to tell. They rely on you to give them a clear and convincing voice.”

A data visual should not tell the whole story but be a prop to be used by the storyteller. Charts support the storyteller. Charts are not the whole story. Charts are a complement for the storyteller by summarizing complex data in a single image.

A picture as a prop has few words. The story (making the argument) is delivered orally or in written prose by the storyteller, the presenter. The data visual in that case should be streamlined to deliver complex data analysis conclusions succinctly to support the presenter, the presentation, the written prose, and the oral arguments, not supplant the presenter. The visual supports the story.

Analysis often produces many views of the data, many graphs. We don't bring them all to a presentation. Out of the many select a few that best support the conclusions of the analysis – that's the story you bring to the audience. These select few graphs have to be compelling. They have to be polished.

The point of this principle is that for a data visual to be most successful the data visual used to support a presentation should be a prop and not be self-contained.

Visual Props - Visuals support the story, they are not the story

- ☐ Is the visual a prop for oral delivery of the story?
- ☐ Have the textual elements on the slide that tell a story been minimized?
- ☐ Is the graph more of a visual prop and less of an infographic?
- ☐ Have the infographic elements been minimized?
- ☐ Has the most important type of analysis graph been selected to support the business story?

Has the visual has been simplified and focused?

Emulate Legendary Storytellers

We have a rich data visualization heritage; we stand on the shoulder of giants, as it were.

They are master storytellers who invented and used data visuals as their props. The question is can you improve your visual by emulating famous chart makers? Are your visuals rooted in their iconic chart exemplars?

For example, see the word of folks like John Snow with his London Cholera Map; Charles Minard and Napoleon's March on Moscow; Hans Rosling who invented Gapminder and moving bubble diagrams; Joseph Priestley and his Chart of Biography and his New Chart of History; and finally, Florence Nightingale and her radar charts of the Crimean War.

The point of this principle is that for a data visual to be most successful we should study and learn how past visualization masters employed the basic charts that they pioneered.

Storytellers - Standing on the shoulder of giants

- ☐ Is the visual improved by emulating famous chart makers?
- ☐ Does the visual have roots in an iconic master chart?

Are past masters and the basic charts that they pioneered emulated?

Chapter 2

Signs



Signs and Sign-Making

Jacques Bertain explains in *Semiology of Graphics* that: *“Data is transformed into graphics to understand. A map, a diagram are documents to be interrogated. But understanding means integrating all of the data. In order to do this it’s necessary to reduce it to a small number of elementary data. This is the objective of the ‘data treatment’ be it graphic or mathematic.”*

We must become good sign makers and send strong unmistakable signals to our audience of the direction they should take in their decision-making.

Give Them a Sign

People need a strong sign to be able to make good decisions. This is where Bertain’s work on Semiotics (the art of sign making) comes in. Are we using signs and symbols properly in our visuals? Does the visual use cultural cues properly? Do we transgress any cultural conventions? Is our audience expecting the symbolism we use in our chart or are they surprised and confused by it?

The science of sign making has three parts: (1) The signifier, the intended meaning; (2) The signified or significant, which is the symbol or icon that stands in for the signifier. for example, a “dog” is represented by a “picture of a dog”; and (3) The sign, the combination that makes up our understanding.

For a data visual to be most successful we must use signs and symbols appropriately.

Signs - Bertain: Semiotics, sign, and signifiers

- ☐ Are signs and symbols used properly in the visual?
- ☐ Are cultural cues used properly in the visual?
- ☐ The visual does not transgress cultural conventions?
- ☐ Is the symbolism in the chart, and that represented by the chart, expected by the viewer?

Is the use of signs and symbols appropriate?

It's Like a Communication System

Sign making for our charts is part of setting up a communication system. We must ask, as with any communication system, does our visual send a strong unmistakable signal? Will the receiver, our audience, be able to decode it? Or is there too much noise and what can we do to reduce it?

In an effective presentation the audience gets your point even in the presence of noise. We encode our message, our story, in signs and symbols of a chart. We communicate it using a transmission system, slides in a presentation, a document, a video, a webcast. The receiver gets our transmission and decodes it: "what does it mean to me, to the numbers I have to

make, to my team, to my company?" Were you clear or did you let noise cloud making your point? Did you increase the signal over the noise?

For a data visual to be most successful we must encode our point clearly so the audience decodes it meaningfully, so we must increase Signal to Noise ratio

Communication - Visualization as a communication system

- ☐ Does the visual send a strong unmistakable signal?
- ☐ Will the receiver be able to decode the signal?
- ☐ Is the noise minimized?

Is the Signal to Noise ratio high?

Design for Function

Our colleague and foremost data journalist Alberto Cairo, in his book *The Functional Art*, urges us to make our charts more functional than artistic. Our charts should inform not entertain.

We should ask ourselves as we create our charts: does the chart inform or entertain? We should avoid having our chart be prized and classified as beautiful art, especially if it fails to inform. We are talking about functionally over making the chart pretty. Avoid frilliness. Have we sacrificed clarity to make our chart pleasing to the eye? We should strive to inform not create chart that appeals to emotions, but rather to reason leading to good decisions.

For a data visual to be most successful make sure the chart is functionally informational rather than beautiful artwork.

Function - Cairo: The Functional Art - Art vs. Clarity

- ☐ Does the chart inform more than entertain?
- ☐ Can the visual be classified as informational only rather than beautiful art?
- ☐ Does information clarity take precedence over making the chart look beautiful?
- ☐ In your judgment is the chart more informational than a piece of art?
- ☐ Does the visual appeal mostly to judgment rather than to the emotions?

Is the chart functionally informational rather than beautiful art?

Chapter 3

Purpose



Design with Purpose

Stephen Kosslyn writes in *Graph Design for the Eye and Mind*: “The usefulness of a graph can be evaluated only in the context of the type of data, the questions the designer wants the readers to answer, and the nature of the audience.”

We must keep in mind who our audience is, what they want and need to know, and what key framed analytical questions we answered to inform them.

Information Need

We use our presentation and its embedded graphs to fulfill the needs of the requester and the organization. We should make sure our visuals fulfill an organizational informational need. We should only be bringing data that is vital to the organization and their mission. The graph must speak to an organizational information need. We should also consider if the requester will be satisfied with this level of information and the news you bring. And most important, does the visual help them make a decision? In other words, does it educate your requester and audience sufficiently to satisfy their needs?

Max Shron, in his powerful *Thinking with Data*, presents a framework for defining data projects, including what data to collect, and how to approach, organize, and analyze the results, and observing patterns of reasoning that help unveil the real problem that needs to be solved.

In his model he explains how we can ascertain what an information need is. Here we will use our common sense and assume the analyst has done the analysis and has clearly determined what the information need is. At the point of communicating results we need to be sure our visuals address that need. Usually the need is about making a decision, often about a gap in the performance of some key performance indicator. Your graph should help the audience have sufficient information to know what to do. We bring our audience information to make an informed decision

For a data visual to be most successful we must make sure the chart fulfills organizational information needs.

Need- Needs of the requester and the organization

- ☐ Does the visual fulfill an organizational information need?
- ☐ Is this data vital to the organization and its mission?
- ☐ Do you think the requester will be satisfied with this level of information and the news it brings?
- ☐ Does the visual help them make a decision?
- ☐ Does the visual educate the requester and the audience and meet their needs?

Does the chart fulfill organizational information needs?

Audience

Together with the audience information needs we should consider all other aspects of our viewers and listeners. What are their biases, what journey are they on, what will they do with the information? Then we must match the visual style to the audience biases, needs and journey. The visual must take into consideration their point of view, and account for their biases, education, training. It should help them with their journey to make their numbers. Any mismatch would introduce noise, perhaps confuse and miss the point you are making.

For a data visual to be most successful it allows for audience biases, needs and journey.

Audience - Cairo: Audience needs, biases, and journey

- ☐ Does the visual style match the audience biases and needs?
- ☐ Does the visual style match the audience journey?
- ☐ Does the visual take into consideration their point of view?
- ☐ Does the visual take into account their biases, education, training?
- ☐ Does the visual help them with their journey?
- ☐ Does the visual help the audience “make their numbers”?

Does the chart allow for audience biases, needs, and journey?

Answer Well Framed Analytical Questions

In the end, to satisfy the information need, we must present our results of answering well-framed analytical questions stemming from those needs. The analytical questions were posed and answered as part of the analysis process. The creation of the communication set of visuals is not the time to discover or search for the answers. We select, out of the many analytical questions we used to inform ourselves of the answers, those few that are the most important. They contain the key evidence, the facts, that support our conclusions. Those key facts must pop out of our visuals as clearly evident. But the basis of the visuals is bringing our audience the very best evidence supporting the conclusions which will lead to action.

For a data visual to be most successful it should be the answer to a key well-framed analytical question.

Frame - Shron: Thinking with Data CoNVO model

- ☐ Does the visual answer a well-framed analytical question stemming from an information need?
- ☐ Is the framed question evident from the visual?
- ☐ After seeing the visual can the viewer express what the framed analytic question behind the visual is?
- ☐ Does the visual’s title directly answer the framed question?

Does the visual answers a well-framed analytical question?

Chapter 4

Perception



Perception

Henry David Thoreau in his *Journal* observes that *“Many an object is not seen, though it falls within the range of our visual ray, because it does not come within the range of our intellectual ray, i.e. we are not looking for it. So, in the largest sense, we find only the world we look for.”* And his key admonition to us as data visualizers is: *“The question is not what you look at, but what you see.”* Looking happens in the eye, seeing happens in the mind.

What are we helping our audience to see?

Use the Eye-Brain System of Seeing

Alberto Cairo in *The Functional Art* writes that the right chart starts in the mind: *“The better we understand the shortcuts the mind uses to make sense of the world, the better we will be able to anticipate them and to take advantage of them for our purposes. As Robert Spence has pointed out visualization is not something that happens on a page or a screen; it happens in the mind. The representation on the page or the screen is merely an aid to facilitate that insight.”*

By using Yarbus’ work we see that the eye is attracted unconsciously to strong focal point images. Therefore, the visual designer should decide the most important things you want your viewers to focus on and make them stand out when they first view chart. They can use Gestalt principles, color theory, employment of the right chart, removing chart junk, and other methods to assist in focusing. You have to guide their viewing so they get the point, almost unconsciously.

For a data visual to be most successful you must make sure that eye of the viewer focuses on the most important point you are trying to make.

Seeing – Anatomy: the human eye-brain system

- ☐ Is the eye of the viewer guided by chart elements to the important points?
- ☐ Does the eye of the viewer focus on the important points made by the visual?
- ☐ Does the viewer's eye go to the most important feature that needs to be stressed?

Does the eye of the viewer focus on the most important point being made?

Employ the Gestalt Principles of Perception

One way to assure that we are guiding the eye to the most important elements of the chart we have designed is to use the principles of the Gestalt Psychology of Perception. Have the principles been used to greatest advantage? The most important principle for our purposes is to make sure the visual has good figure/ground differences. Does the main point visually stand out? Then we can consider secondary effects such as asking if the grouping has been used to best effect. Has connectedness been used effectively? Has flow been used properly?

For a data visual to be most successful careful use of the principles of the Gestalt psychology of perception should be employed when creating the visual.

Design with Quality

The architect Christopher Alexander has a unique perspective on quality, especially when it comes to designing something that must be useful to others. We shall consider our efforts at building charts as such and endeavor to be of use in informing others. On that premise we can apply his formalism.

He tells us that a design can have a timeless quality, which is hard to define, but you see it and you feel it when it's there. He calls it the quality without a name. For a building it is something that you are comfortable living and being inside of, it is functional but also has its own beauty, and fulfills a human need.

The human need in the matter of data visuals is to be informed. A visual will have that quality of being alive, and being very useful to human beings if we use the principles Alexander outlines.

He tells us that the key element is to use a pattern language, which is a collection of techniques that produce quality visuals—essentially, all the good principles we outline here. Patterns resolve forces in tension and, in our case, bring knowledge. The tension in our case is between knowing and not knowing. So the question we must answer positively is: does our visual resolve the tension in the viewer between not knowing (before seeing our visual) and knowing (seeing our visual) and having their need satisfied. Is their ignorance dispelled?

In other words, our visuals must be "alive." To be "alive," a visual must provide a service. The service is the resolution of a tension the viewer brings to the chart. They want to know. If the chart informs, the tension is relieved, the viewer now "knows." If the viewer is still puzzled over the information after viewing, the chart does not have the living quality of informing and releasing the tension of ignorance.

For our chart to be effective, did it inform, did it relieve tension? Then we can say it is a quality chart.

Mind - Gestalt: psychology of perception

- ☐ Are basic Gestalt principles of perceptions used to the best advantage?
- ☐ Does the visual have good figure/ground differences?
- ☐ Has grouping been used effectively?
- ☐ Has connectedness been used effectively?
- ☐ Has flow been used effectively?

Have the principles of the Gestalt psychology of perception been thoughtfully employed in the visual?

Exercise 5

Method



Method

We must use color appropriately, make sure our charts are devoid of all visual noise, and make sure the PowerPoint slide has a title that gets to the point of your message.

Use Color Effectively

Color should be used judiciously. It really enhances the chart. But if you use Excel, for example, the program chooses which colors to display based on some internal formula. That does not always lead the best color combination.

First you must use color sparsely. Most of your chart should be in black and white. Color should be left for those elements you want to draw to the viewer's attention. For example, you could gray out the axis somewhat to make it fade into the background. Does your use of color grab the viewer's attention?

Color should also be used semantically correct. For example, is your data line for the US as blue and for China as red, or in reverse? One is correct and the other makes the viewer stop and pause to wonder what you mean by having it in a different color scheme than expected. That makes the viewers waste time in absorbing your meaning and you may even lose them.

Also be aware that some people suffer from color dysfunctions in their eyesight (color blindness, for example) and your color scheme should account for that.

For a business data visual to be most successful color should be used judiciously and sparsely.

Color - Use of color, dysfunctions

- ☐ Has color been used judiciously and sparsely?
- ☐ Is the use color semantically correct?
- ☐ Does the use of color grab the viewer's attention?
- ☐ Is color used to differentiate crossing elements?
- ☐ Is color used in area charts sparsely?
- ☐ Is color used to highlight results in tables sparsely?

Is color used judiciously and sparsely

Remove all Chart Junk

Richard Tufte is a prime mover in helping us understand that chart junk detracts from viewer's comprehension of our charts. Just as we must declutter when we prepare our family home to be sold clear up the viewing space in a visual as much as possible. Be ruthless. Use the redo and undo function in your chart maker. Put the feature in and then delete it. Go back and forth putting it in, and taking it out, to determine if valuable information would be lost if it's not there. Is there simpler way to make that point?

Tufte use the term chartjunk for unnecessary or confusing visual elements in charts and graphs. Any markings and visual elements can be called chartjunk if they are not part of the minimum set of visuals necessary to communicate the information understandably.

For a data visual to be most successful the visual should be clear of unnecessary visual elements that do not help the viewer understand.

Chart Junk - Tufte: Chartjunk

- ☐ Has all chart junk been removed?
- ☐ Has the data to ink ratio been properly balanced?
- ☐ Have all unnecessary elements been removed, including grid elements?
- ☐ Have all unnecessary elements been removed, including extra zeroes in the axis labels?
- ☐ Have all unnecessary extra bars been removed (could you have used a bullet chart format)?
- ☐ Have unnecessary pie chart sections been removed (not more than 5)?
- ☐ Has area coloring that detracts been removed?
- ☐ Have unnecessary icons and other images been removed?
- ☐ Does the chart have sufficient drama?

Is the visual is clear of unnecessary visual elements not leading to a clear point being made?

Tell the Story with the Title

Our friends at McKinsey and Company, the world-famous consulting powerhouse, have a wonderful practice in titling their presentation slides. The title of the slide (or even the visual, but not both) makes the business point of the graph explicitly. It tells them what the graph means right in the title. They don't let the viewers try to figure it out for themselves. They may draw the incorrect conclusion. You need to tell them, right there, at the top of your slide. Be succinct but express it as a complete thought, a phrase will suffice.

Charts should also have direct labeling for series. Avoid using a legend that makes the viewer's eye go back and forth and get lost as they try to get the point. Tell them directly what each chart feature is.

For a data visual to be most successful the title of the chart should convey the point you are trying to make by using the chart.

Title – McKinsey: A Better Way to Title Charts

- ☐ Does the title of the chart make the point directly?
- ☐ Does the title of the chart tell the story?
- ☐ Does the title of the chart answer the question being raised?
- ☐ Is the McKinsey method of titling used?

Does the title of the chart convey the point being made with the chart?

Exercise 6

Charts



Charts

In this last lab we will experiment with selecting the right chart to match our purpose and story. We will also optimize tables so they can be used effectively in the short time they will be displayed during a presentation.

Use the Right Chart

The researchers Cleveland and McGill have developed a useful scale for elementary perceptual tasks which we apply to decide the type of graph that should be used for the level of accuracy desired. Cleveland and McGill's functional scale tells us that the higher the encoding method, the more accurate the comparison it facilitates.

This scale should guide us to match the type of graph to use for the question being answering. Using this scale, we can ask: (1) Is the right chart being used for the intended purpose? (2) Does the visual use the right level of encoding for the level of accurate judgment desired? (3) Does the chart type match the type of question being answered?

For a data visual to be most successful the type of chart used matches the level of judgement required.

Right Chart - Cleveland and McGill: Functional Scale

- ☐ Is the right chart being used for the intended purpose?
- ☐ Does the visual use the right level of encoding for the level of accurate judgment desired?
- ☐ Does the chart type match the type of question being answered?

Does the type of chart being used match the level of judgment required?

Select the Chart Type Effectively

For this lab we should consider answering a few basic questions in the use of the chart type that was selected: (1) are the charts answering the right business question?; (2) does the chart match the business question being presented?; and (3) have the basic four chart types been used properly?

Each chart type is best suited to presenting answers to certain types of questions. For example, a pareto chart is great for showing the 80/20 percent contributions for certain contributors. Pie charts are best for contributions to a whole. Match the point you are trying to make to type of chart that suits that type of information best.

For a data visual to be most successful the chart type used matches the business question being answered.

Selection – The Basic Charts

- ☐ Are your charts answering the right business question?

- ☐ Does the chart match the business question being presented?
- ☐ Have the basic four chart types been used properly?

Does the chart type used match the business question being answered?

Enhance Table Data for Emphasis

In her a clear blog, Cole Nussbaumer-Knaflic demonstrates how a focus on communication leads from a formatted table to a more intuitive view of key characteristics and elements within the dataset.

When using tables in our visuals we must ask: are we using the table to analyze or to tell? Is the use of referenceable or glanceable visual appropriate? Also we must ask if the referenceable visual been designed to be legible and readable? Do the tables have enough white space? Has shading of the table been used appropriately and sparingly? Has the emphasis been appropriately added with conditional formatting? And have thumbnail graphs such as Sparklines been employed to add insight?

For a data visual to be most successful have referenceable visuals (tables) made readable with appropriate conditional formatting and thumbnail graphs.

Tables - Analyze or Tell?

- ☐ Is the use of a table appropriate in this case?
- ☐ Is the table readable?
- ☐ Do the tables have enough white space?
- ☐ Is shading of the table used appropriately?
- ☐ Has appropriate emphasis been added to the table with conditional formatting?
- ☐ Have thumbnail graphs such as Sparklines been employed to add insight?

Are referenceable visuals (tables) readable with appropriate conditional formatting and thumbnail graphs used for emphasis?

References



References

- Alexander, Christopher. *The timeless way of building*. Vol. 1. New York: Oxford University Press, (1979).
- Argyris, Chris, and Donald A. Schon. *Theory in practice: Increasing professional effectiveness*. Jossey-Bass, 1974.
- Bertin, Jacques. "General theory, from semiology of graphics." *The Map Reader: Theories of Mapping Practice and Cartographic Representation* (2011): 8-16.
- Cairo, Alberto. *The Functional Art: An introduction to information graphics and visualization*. New Riders, 2012.
- Cleveland, William S., and Robert McGill. "Graphical perception: The visual decoding of quantitative information on graphical displays of data." *Journal of the Royal Statistical Society. Series A (General)* (1987): 192-229.
- Cotgreave, Andy, "The 5 Most Influential Data Visualizations of All Time", Tableau, 2013, <https://www.tableau.com/learn/webinars/5-most-influential-visualizations-all-time>.
- Few, Stephen. *Now you see it: simple visualization techniques for quantitative analysis*. Analytics Press, 2009.
- Kosslyn, Stephen M. *Graph design for the eye and mind*. OUP USA, 2006.
- Lewis, Richard. *When cultures collide*. Nicholas Brealey Publishing, 2010.
- Medina, J. (2012). *Brain Rules*. Pear Press, Seattle, 2012.
- Nussbaumer Knafllic, C., Story Telling with Data: Visualizing Opportunity", in September 16, 2015, Makeovers, <http://www.storytellingwithdata.com/blog/2015/9/16/visualizing-opportunity>.
- Rosling, Hans. "Hans Rosling shows the best stats you've ever seen, 2006." *Internet*: http://www.ted.com/talks/hans_rosling_shows_the_best_stats_you_ve_ever_seen.html.
- Shron, Max. *Thinking with Data: How to Turn Information into Insights*. O'Reilly Media, Inc., 2014.
- Thoreau, Henry David. *The Writings of Henry David Thoreau (Volume V)*. (1884).
- Tufte, Edward, and P. Graves-Morris. "The visual display of quantitative information; 1983." (2014).
- Yarbus, Alfred L. "Eye movements during perception of complex objects." *Eye movements and vision*, pp. 171-211. Springer, Boston, MA, 1967.